

7.4.25

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Question: The locus of the centre of a circle, which touches the circle is $x^2 + y^2 - 6x - 6y + 14 = 0$ and also touches the y -axis, is given by the equation:

- 1) $x^2 - 6x - 10y + 14 = 0$
- 2) $x^2 - 10x - 6y + 14 = 0$
- 3) $y^2 - 6x - 10y + 14 = 0$
- 4) $y^2 - 10x - 6y + 14 = 0$

Solution:

Given circle equation can be represented as

$$\|\mathbf{x}\|^2 + 2 \begin{pmatrix} -3 \\ -3 \end{pmatrix}^T \mathbf{x} + 14 = 0 \quad (4.1)$$

The centre of circle is $\mathbf{c}_1 = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$ and radius $r = 2 \left(\because f = 14, \quad \mathbf{u} = \begin{pmatrix} -3 \\ -3 \end{pmatrix} \right)$

Let the centre of the moving circle be $\mathbf{c} = \begin{pmatrix} h \\ k \end{pmatrix}$

As the circle touches Y -axis, Distance of a point from y -axis is given by

$$\text{radius } R = h$$

The equation of moving circle is

$$\|\mathbf{x}\|^2 + 2 \begin{pmatrix} -h \\ -k \end{pmatrix}^T \mathbf{x} + k^2 = 0 \quad (4.2)$$

From (4.1) and (4.2), the intersection of curves

$$\|\mathbf{x}\|^2 + 2 \begin{pmatrix} -3 \\ -3 \end{pmatrix}^T \mathbf{x} + 14 - \|\mathbf{x}\|^2 - 2 \begin{pmatrix} -h \\ -k \end{pmatrix}^T \mathbf{x} - k^2 = 0 \quad (4.3)$$

$$2 \begin{pmatrix} h-3 \\ k-3 \end{pmatrix}^T \mathbf{x} = k^2 - 14 \quad (4.4)$$

The equation of the common tangent is

$$\begin{pmatrix} h-3 \\ k-3 \end{pmatrix}^T \mathbf{x} = \frac{k^2 - 14}{2} \quad (4.5)$$

Distance between the common tangent and centre of given circle equal to it's radius

$$2 = \frac{\left| \begin{pmatrix} h-3 & k-3 \end{pmatrix} \begin{pmatrix} 3 \\ 3 \end{pmatrix} - \frac{k^2-14}{2} \right|}{\sqrt{(h-3)^2 + (k-3)^2}} \quad (4.6)$$

$$2 = \frac{\left| 3(h-3) + 3(k-3) - \frac{k^2-14}{2} \right|}{\sqrt{(h-3)^2 + (k-3)^2}} \quad (4.7)$$

$$2\sqrt{(h-3)^2 + (k-3)^2} = 3h + 3k - \frac{k^2}{2} - 11 \quad (4.8)$$

$$(4.9)$$

Squaring both sides

$$4((h-3)^2 + (k-3)^2) = (3h + 3k - \frac{k^2}{2} - 11)^2 \quad (4.10)$$

$$4h^2 + 4k^2 - 24h - 24k + 72 = \frac{k^4}{4} + 9h^2 + 20k^2 + 121 + 18hk - 66h - 66k - 3hk^2 - 3k^3 \quad (4.11)$$

$$k^4 - 12k^3 + (64 - 12h)k^2 + 72hk + 20h^2 - 168h - 168k + 196 = 0 \quad (4.12)$$

$$(k^2 - 10h - 6k + 14)(k^2 - 6k - 2h + 14) = 0 \quad (4.13)$$

The possible general loci are

$$y^2 - 10x - 6y + 14 = 0 \quad (4.14)$$

$$y^2 - 6y - 2x + 14 = 0 \quad (4.15)$$

From the options , the locus of the centre of the circle is

$$y^2 - 10x - 6y + 14 = 0 \quad (4.16)$$

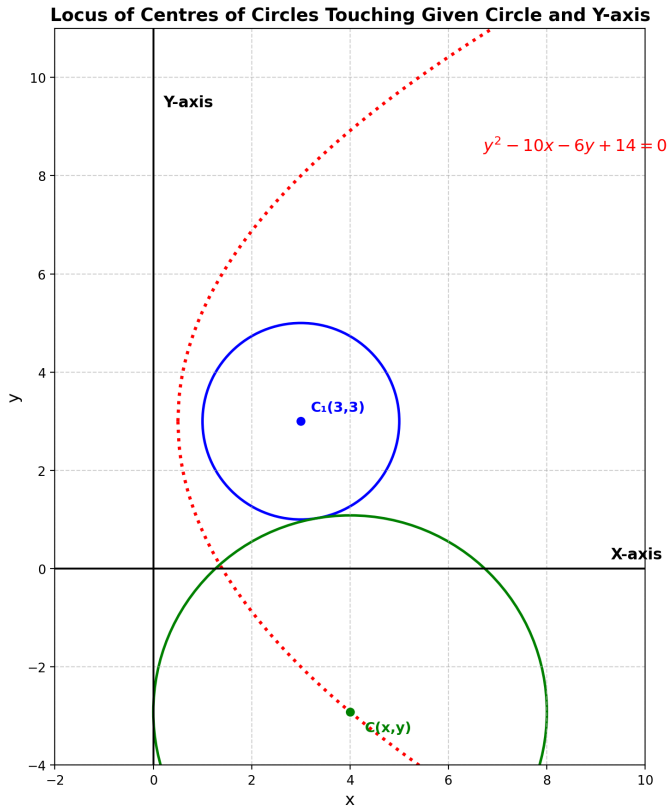


Fig. 4.1